AES S1 - Basic Electronics

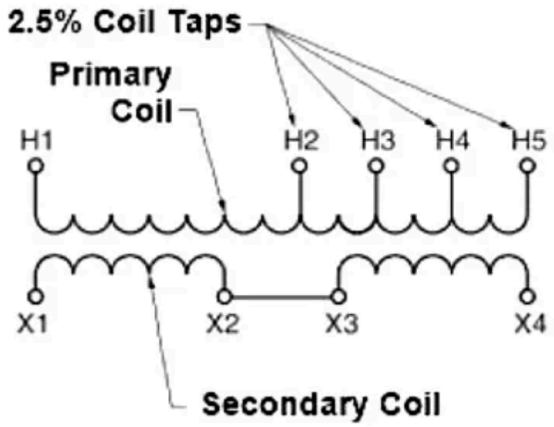
2026-27

Universitat Pompeu Fabra Barcelona

1. We want to design an AC circuit with two inductors, each one with an impedance of 16Ω.
    a. Calculate the equivalent impedance if we connect them in series
    b. Calculate the equivalent impedance if we connect them in parallel
2. We have a voltage divider circuit with two resistors connected in series, R_1=1.5kΩ and R_2=2.7kΩ.
    a. Calculate the output voltage in R_2 for an input voltage of 1V.
    b. Estimate the electric power dissipated in R_2
3. Calculate the voltage generated at the output of a power amplifier that generates 100W,
    a. when connected to an 8Ω loudspeaker.
    b. when connected to a 4Ω loudspeaker.
4. Estimate the electric power (W) at the output of a power amplifier that provides 80V,
    a. when connected to two 4Ω loudspeakers with series interconnection.
    b. when connected to two 4Ω loudspeakers with parallel interconnection.
5. Calculate the capacitance needed in a 1st order RC high pass filter implemented with a resistance of 1kΩ to obtain a cut-off frequency of 80Hz.
6. Calculate the resonance frequency of a series RLC notch filter implemented with a resistance of 100Ω, a coil of 100mH and a condenser of 10μF.
7. Given a 1_{st} order RC band pass filter for f_{HPF}=800Hz and f_{LPF}=1.6kHz.
    a. Estimate R and C values.
    b. Provide the response velocity (tau) for each filter.
    c. Which is the slope (in dB/oct) for each filter.
    d. Design the 2_{nd} order BPF for the same cutoffs.
    e. Draw the circuit with all elements and values.
    f. Provide the slope and the phase shift in the cutoffs in the 2_{nd} order BPF.
8. Considering a 10dB L-pad attenuator and V_{IN}=0.707V
    a. Which is the output level in dB_V?
    b. Which is the output voltage?
    c. Repeat a and b for a 24dB L-pad attenuator.
    d. Estimate the resistance values for a 24dB L-pad attenuator for Z=600Ω.
9. Considering a passive 2_{nd} order LC crossover in a 2-way loudspeaker. Estimate the cutoff frequency assuming the following values:
    a. L_{HPF} = 0.25mH and C_{HPF} = 21.11μF
    b. L_{LPF} = 1.2mH and C_{LPF} = 16μF

10. Design a 2nd order LC crossover in a 2-way loudspeaker with 1.6kHz cutoff frequency. Estimate the LC values for each filter and provide the complete circuit.

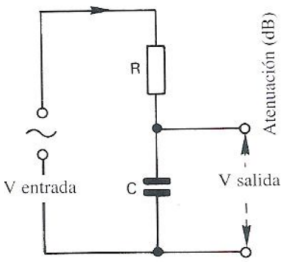
11. Considering a transformer with 1:3 ratio and V_p=0.707V.
    a. Estimate Vs.
    b. Propose a setting for the N_p and N_s
    c. Imagine we have to design a transformer with multi-tap coils with N_p=1720 turns and V_p=0.707V. If the V is 10% low in the second tap, which would be the corresponding number of turns? Note: V/turn = V_p / N_p



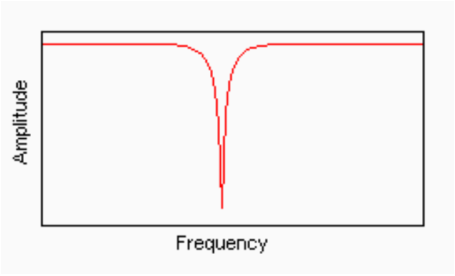
    d. Make the same for 2.5%, 5% and 20%.
    e. If Z_p / Z_s = (N_p / N_s)², compute the impedance ratio for 1:2 transformer.
    f. Which is the value of Z_s for Z_p = 600Ω

12. Find out the matchings:

a)



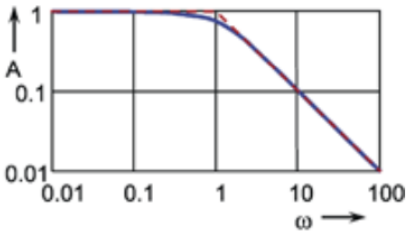
1)



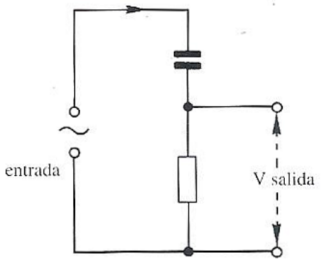
b)

$$f_c = \frac{1}{2\pi RC}$$

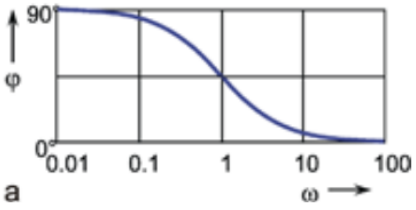
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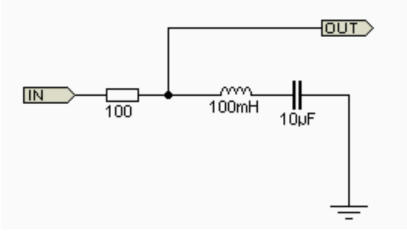
c)



3)



d)



4)

